SkyPass: Exact Weather-Aware Scheduling of Satellite Passes for Small Ground Stations

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***Abstract*—Observation planning at a small ground station is presently distributed across unconnected tools: one predicts passes, another evaluates optical visibility, cloud conditions are checked manually, and conflicts between overlapping passes are resolved by the operator. Cloud cover is the least well handled of these, since conventional predictors enumerate passes without reference to it. This paper presents SkyPass, an integrated planner that carries a catalogue of orbital elements through propagation, pass extraction, visibility scoring, cloud-forecast integration and exact conflict resolution to an executable timetable. Pass extraction uses coarse bracketing with bisection refinement, achieving 0.055 s mean agreement with an independent implementation while requiring 23.7× fewer propagator evaluations than a dense scan. Conflict resolution is formulated as weighted interval scheduling under a nightly observing budget and solved exactly by a dynamic program whose running time grows linearly with the number of candidate passes and the nightly limit; it matched exhaustive search on all 2000 randomised instances tested. Forecast skill is verified against ERA5 reanalysis at seven stations and scheduling value at six, over 60 days. The principal finding is that forecast value is determined by the decision the forecast informs: a station required to observe every night obtains no benefit, whereas one able to select its observing nights improves realised yield by 19.0% (expected value) and 37.6% (clear-sky count). Calibrating the forecast against reanalysis raises the gain in every regime tested; under a nightly quota the uncalibrated forecast performs worse than disregarding weather.**

***Index Terms*—satellite pass prediction, SGP4, ground station scheduling, interval scheduling, forecast calibration, optical observation**

I. Introduction

Small ground stations, such as university laboratories, amateur-radio clubs and individual optical observers, face a recurring planning problem. Several hundred satellites are potentially observable and over a week generate thousands of passes, of which only a few dozen merit the operator's time, and the station can attend to one object at a time.

Existing tools address only parts of this problem. Pass predictors compute rise and set times from two-line elements (TLEs) and the SGP4 propagator [1], [2], and some add an illumination test so that optical observers are not shown daylight passes. A few applications display a weather indication beside a pass, but to our knowledge none allow the forecast to *rank* passes or drive a scheduling decision, and none resolve conflicts between overlapping passes.

The treatment of weather is the more consequential omission. At the site used as a running example (Chennai, India, 12.92° N), the mean observed cloud cover over optically viable passes was 88%, under which most geometrically favourable passes are not observable. A predictor ranking passes by culmination elevation will nonetheless recommend an overcast night. The scheduling omission is less apparent. Once several satellites are tracked their passes overlap, and a single-aperture station must select among them. The literature addressing this problem does so from the operator side of the link, as spacecraft tasking and satellite range scheduling, formulated as mathematical programs or addressed with metaheuristics and learning methods [3]–[7]. Such formulations impose constraints a small station does not have, and solver dependencies it cannot justify.

This paper shows that the small-station problem admits a substantially simpler exact formulation, and that the treatment of weather is what improves realised outcomes. The contributions are: an integrated pipeline accounting for the passes removed by each physical constraint; a validated pass extractor; exact capacity-constrained conflict resolution requiring neither a solver nor a metaheuristic; a weather claim verified against ERA5 reanalysis [8]; two negative results, namely that weather-awareness yields no benefit under a fixed nightly quota and that the uncalibrated forecast degrades performance there; and a measured element-set error budget over 33 223 aged comparisons.

II. Related Work

SGP4 and its two-line element sets remain the standard basis for rapid satellite position prediction [1], [2]. Accuracy is limited by the element sets rather than the propagator: TLE-derived positions carry errors that grow with element-set age, as measured against on-board GPS [9] and estimated statistically [10].

Whether a satellite is *seen* depends on illumination and photometry, not only geometry. Standard treatments use a conical Earth-shadow model [1] and a diffuse-sphere phase

function for apparent magnitude [11], and recent megaconstellation brightness work has refined these [12], [13].

Satellite range scheduling is an oversubscribed combinatorial problem [3], usually approached with multi-objective optimisation, metaheuristics or reinforcement learning [5], [14]. Earth observation scheduling is a large adjacent literature [7], [15], in which cloud cover appears as an uncertainty to be hedged during imaging tasking [16], [17], although always on the *operator* side. Where the underlying structure is single-machine interval selection, exact polynomial algorithms exist: weighted interval scheduling is a textbook dynamic program [18], and harder members of the family continue to attract exact methods [19].

Professional observatories schedule around weather using queue schedulers that re-plan as conditions change [20], [21], but assume an institutional software stack and on-site sky monitoring. Forecast verification and forecast value are mature fields [22], [23], and statistical post-processing is essential before a forecast drives a decision [24]. To our knowledge this machinery has not been applied to observer-side pass planning.

III. System Design

SkyPass is a five-stage pipeline (Fig. 1). Each stage is a pure function of the preceding one, so any stage may be ablated independently, which is the mechanism by which the weather claim is evaluated.

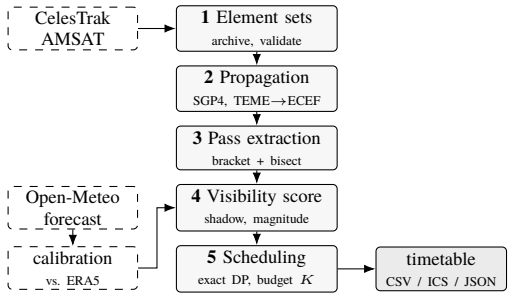


Fig. 1. SkyPass pipeline. The cloud forecast is calibrated against reanalysis before it is allowed to influence the score; the scheduler then resolves conflicts exactly under the station's nightly observing budget.

A. Element sets, propagation and pass extraction

Element sets are fetched from CelesTrak [25], checksum-validated, deduplicated per NORAD identifier keeping the newest epoch, and *archived under the date of retrieval*: CelesTrak serves only current elements, so without a local archive a published prediction cannot be repeated. Sets whose epoch age exceeds $\tau_{\max}$ are rejected.

Positions come from the reference SGP4 implementation. The TEME vector is rotated to ECEF by Greenwich mean sidereal time and converted to topocentric azimuth, elevation and range on the WGS-84 ellipsoid. Polar motion is neglected, being below 15 m at the surface and far under the SGP4 error budget. Atmospheric refraction [26] is reported but not applied to the horizon mask, so results do not depend on assumed surface conditions.

Because cost is dominated by propagator calls, pass extraction is designed to minimise them (Algorithm 1). Elevation is sampled on a coarse grid to *bracket* horizon crossings; within a pass elevation is smooth and unimodal, so a sign change of $\varepsilon(t) - \varepsilon_{\text{mask}}$ between adjacent samples brackets exactly one crossing. Each bracket is refined by bisection to 0.1 s, and culmination by golden-section search. The coarse step is $P/240$ bounded to 5 s to 30 s, so short-period orbits, whose passes are shortest, are sampled most finely.

Algorithm 1 Pass extraction for one object

Require: element set σ, window $[t_0, t_1)$, mask ε_m, tolerance δ

$h \leftarrow \max(5, \min(\Delta_{\max}, P(\sigma)/240))$
$t \leftarrow t_0$; $e_{\text{prev}} \leftarrow \varepsilon(\sigma, t)$; $a \leftarrow$ **nil**
while $t < t_1$ **do**
  $t' \leftarrow \min(t + h,\, t_1)$; $e \leftarrow \varepsilon(\sigma, t')$
  if $e_{\text{prev}} < \varepsilon_m \le e$ **then**
    $a \leftarrow \textsc{Bisect}(t, t', \varepsilon_m, \text{rising}, \delta)$
  else if $e_{\text{prev}} \ge \varepsilon_m > e$ **and** $a \neq$ **nil then**
    $l \leftarrow \textsc{Bisect}(t, t', \varepsilon_m, \text{setting}, \delta)$
    $c \leftarrow \textsc{GoldenMax}(a, l, \delta)$
    emit pass (a, c, l); $a \leftarrow$ **nil**
  end if
  $t \leftarrow t'$; $e_{\text{prev}} \leftarrow e$
end while

B. Visibility scoring

Each pass receives a score $S \in [0, 1]$ combining a geometric quality term with multiplicative gates:

$$S = \underbrace{(w_e f_e + w_d f_d)}_{\text{geometry}} \cdot f_m \cdot f_s, \tag{1}$$

where $f_e = \min(\varepsilon_{\max}/60^\circ, 1)$ and $f_d = \min(T/480\,\text{s}, 1)$ saturate culmination elevation and duration, f_m is a photometric factor and f_s a sky-condition factor.

The multiplicative form is deliberate: an eclipsed pass or one in a bright sky scores zero irrespective of its geometry, and expected cloud scales the score down in proportion, which an additive score cannot represent. For optical work the object must be sunlit and the observer in darkness; Earth shadow uses the dual-cone construction [1], and apparent magnitude follows the standard diffuse-sphere relation [11], [27]

$$m = m_{\text{std}} + 5\log_{10}\frac{d}{1000\,\text{km}} - 2.5\log_{10} F(\psi), \tag{2}$$

with $F(\psi) \propto (\pi - \psi)\cos\psi + \sin\psi$ normalised to $F(0) = 1$. In radio mode the illumination and photometric terms are dropped.

C. Weather integration and calibration

The sky factor is $f_s = 1 - \hat{c}$, where $\hat{c}$ is the expected cloud fraction at culmination, interpolated linearly between hourly samples. Crucially $\hat{c}$ is *not* the raw forecast, which is heavily over-confident as a predictor. We fit

$$\hat{c} = \mathbb{E}[c \mid c_f] = a + b\, c_f \tag{3}$$

by least squares on a window that *ends before* the period being planned, restricted to local night hours, and clip $\hat{c}$ to $[0, 1]$. The slope b quantifies the strength of the signal: $b \approx 1$ indicates that the forecast may be used directly, whereas $b \approx 0$ indicates that it carries little information, in which case the calibrated value converges toward site climatology, the appropriate fallback. For the threshold value model the planner instead weights each pass by the calibrated probability of clear sky, $P(c \leq \theta \mid c_f)$, estimated in ten forecast bins on the same window and shrunk toward the site's base rate with 20 pseudo-observations, so each value model is paired with the planner that maximises its expectation.

D. Conflict resolution

A single-aperture station observes one object at a time and requires a setup gap g between targets, so selecting the best mutually compatible subset of passes is weighted interval scheduling. Writing $v_j = S_j \pi_j$ for the value of pass j (score times mission priority), indexing passes $1, \ldots, n$ by finish time and letting $p(j)$ be the largest $i < j$ compatible with j (0 if none), the textbook recurrence [18] gives the exact optimum in $O(n \log n)$.

This formulation alone is insufficient: left unconstrained, a station observes every candidate, so weather-awareness has no selection to perform. Operational stations are limited by operator and instrument time, so we impose a capacity budget of K observations per night, extending the recurrence with a cardinality dimension:

$$\mathrm{OPT}[j][k] = \max\big(\mathrm{OPT}[j-1][k],\ v_j + \mathrm{OPT}[p(j)][k-1]\big), \quad (4)$$

with $\mathrm{OPT}[0][k] = \mathrm{OPT}[j][0] = 0$. This is exact. The table has one entry for each of the n candidate passes and each count up to the nightly limit K, and each entry is filled in constant time, so after the sort the cost is $O(nK)$: doubling either the candidates or the limit doubles the work. Observing nights are disjoint in time, so applying the budget per night and solving each night independently remains exact. Nights are defined in local solar time so that an evening and the small hours that follow belong to the same session.

IV. Experimental Setup

The catalogue comprises 635 unique objects from CelesTrak GP groups, checksum-validated and deduplicated per object. Seven stations span distinct cloud climatologies: Chennai and Bengaluru (monsoon), Leh (high-altitude arid), Tucson (desert), Singapore (equatorial), and Reykjavík and Svalbard (maritime polar). Unless stated otherwise, the mask is 10°, the dark-sky threshold −6°, and the setup gap 5 min.

All three weather products come from Open-Meteo [28]: the operational forecast; archived past forecasts at lead times of 1–7 days; and ERA5 reanalysis [8] as verification truth. The archive endpoint has to be pinned to the ERA5 model: its default best-match series *is*, for recent dates, the archived forecast, and verifying against it reports a spurious skill of 1.0.

Timing is checked against a brute-force 1 s dense scan and against Skyfield [29], which shares only the certified SGP4 core and implements time scales, frames and event finding independently. All runs are single-threaded, under Python 3.13.5 with `sgp4` 2.27 on a commodity laptop.

TABLE I
Pass-timing agreement over the full catalogue (635 objects, 1761 matched passes, 24 h horizon). Values in seconds.

	vs. 1 s dense scan		vs. Skyfield	
Quantity	mean	p95	mean	p95
$\lvert\Delta t_{\mathrm{AOS}}\rvert$	0.504	0.954	0.060	0.157
$\lvert\Delta t_{\mathrm{TCA}}\rvert$	0.248	0.476	0.055	0.110
$\lvert\Delta t_{\mathrm{LOS}}\rvert$	0.502	0.956	0.059	0.156
$\lvert\Delta \varepsilon_{\max}\rvert$ (deg)	0.0008	0.0022	0.0034	0.0102

V. Results

A. Pass-prediction accuracy and cost

Table I compares SkyPass against both references over 635 objects and 1761 matched passes.

Against Skyfield, mean agreement is 0.060 s on acquisition and 0.055 s on culmination. The apparent 0.504 s "deviation" from the dense scan is not SkyPass error: a 1 s grid locates a crossing only to within its own step, so it lags the true crossing by a mean of 0.5 s. The refined estimator is thus more accurate than the brute-force reference it is conventionally compared against. The cost difference is large: 2 311 148 propagator evaluations against 54 864 000, a factor of 23.7. Detection recall is 99.78%, the residual being very short grazing transits that peak barely above the mask.

B. Element-set limits on prediction accuracy

Sub-second numerical agreement is meaningful only if the orbital elements support that precision. In the operational catalogue the median element-set age was 0.42 days (90.2% under one day). CelesTrak and AMSAT element sets differ by a mean of 0.041 s in culmination over 308 matched passes, but the median difference is 0.00 s: the two providers largely redistribute the same upstream sets, so this bounds redistribution differences rather than independent orbit determination.

To measure decay with staleness we obtained historical element sets for 60 low-Earth objects, took the freshest set before a target night as reference, and re-predicted the same passes from progressively older sets, giving 33 223 aged comparisons (Fig. 2).

Within a day of epoch the elements contribute 0.05 s of culmination error, the same order as the 0.055 s numerical agreement of Section V-A, so neither term dominates at that age. Degradation is then sharply super-linear and heavy-tailed: a mean of 4.3 s (median 0.70 s) at 7–14 days, 16.2 s at 14–30, and 75.3 s (median 13.69 s) beyond a month. The design consequence is that a 5–7 day staleness cutoff is appropriate for timing-critical work, beyond which the elements rather than the algorithm set achievable accuracy, and that daily element refresh is the single largest lever on prediction accuracy.

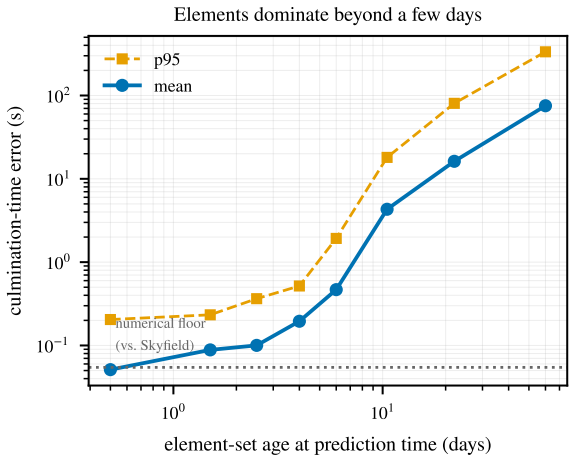


Fig. 2. Culmination-time error against element-set age, over 33 223 aged comparisons. The dotted line is the numerical agreement with Skyfield. Fresh elements land on that floor; by a week the mean is nearly two orders of magnitude above it.

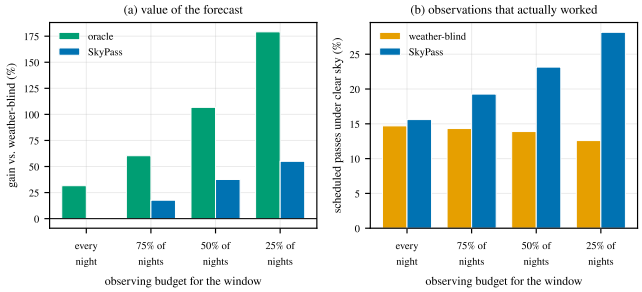


Fig. 3. Weather-awareness is worth nothing to a station obliged to observe every night, and a great deal to one that may choose its nights. (a) Gain over the weather-blind exact planner, threshold value model; the oracle plans with the observed cloud. (b) Share of scheduled passes under clear sky.

C. Skill of the cloud forecast

Weather-aware planning is justified only if the forecast is informative at the lead times over which an observer plans. Archived forecasts were verified against ERA5 across all seven stations over 60 days, scoring the operational binary decision of whether a given hour is sufficiently clear to observe at a 30% cloud threshold. Accuracy alone is misleading when clear hours are rare, so we report the Heidke skill score, a chance-corrected measure in wide use for categorical forecast verification [30].

Skill decays steadily with lead time, from HSS 0.347 at one day to 0.289 at three days and 0.114 at seven. The one-day forecast outperforms both naive baselines (persistence 0.206, climatology 0 by construction), but by five to seven days it no longer does. A weather-aware planner should therefore run on a rolling short horizon and re-plan daily. Skill is also strongly site-dependent, the spread between stations being comparable to the decay with lead time and lowest at the two Indian monsoon stations, Chennai reaching zero skill beyond four days. The expected benefit of weather-awareness is thus a local property rather than a universal constant.

D. Value of weather-aware scheduling

Five planners select from an identical candidate set, drawn from 300 objects, at each of six stations over 60 days (Svalbard has no observable passes), and every timetable is scored against ERA5. Since the valuation of a clouded observation is a modelling choice, we report both an *expected value* model, $\sum_j v_j(1-c_j)$, and a *threshold* model, $\sum_j v_j \mathbb{1}[c_j \le \theta]$ with $\theta = 30\%$.

Under a fixed nightly quota the forecast yields no benefit: SkyPass gains -1.5% under the expected-value model and -0.2% under the threshold model, against oracle headroom of 12.0% and 31.6% respectively, and the station-bootstrap intervals of both gains include zero. This is not because the forecast is uninformative: at the candidate-pass timestamps, forecast and observed cloud correlate at $r = 0.41$, and the clearest-forecast decile does identify clearer skies, reducing observed cloud by 0.209.

The limitation is that a nightly quota permits the planner only to *reorder passes within a night*, which two measured properties render ineffective. Cloud is strongly autocorrelated over the few hours a session spans, the between-night standard deviation of observed cloud being $2.2\times$ the within-night value; and pass quality varies more than sky conditions do, the ninetieth-percentile pass carrying $1.8\times$ the mean weather-free value. Since the score is a product, selecting for clear sky requires surrendering base value, an exchange that is unfavourable where quality is the more variable factor.

Given the same nightly cap together with a budget for the window, so that it may *skip* a night, the result reverses (Fig. 3): the choice of which nights to use is a day-scale decision, the scale at which the forecast retains skill. Under the budget permitting observation on 50% of nights (90 observations across the window), SkyPass improves realised yield by 19.0% under the expected-value model and 37.6% under the threshold model (95% intervals 9.2–24.9% and 3.1–80.4%), recovering 41% and 35% of the attainable oracle gains. The fraction of scheduled passes falling under clear sky rises from 13.9% to 23.1%, and tightening the budget to a quarter of the nights raises the threshold-model gain to 55.0%.

Calibration raises the gain in every regime. Under the quota, supplying the model cloud field directly to the score costs -9.2% relative to disregarding weather; under the half budget the raw field still helps (8.0% and 29.9%) but less than the calibrated one. Fig. 4 indicates the cause: fitted slopes lie between 0.17 and 0.61 and intercepts between 0.07 and 0.70, so clear forecasts are followed by more cloud than predicted and cloudy ones by less. Calibration (3) shrinks the forecast toward climatology in proportion to its demonstrated reliability.

Repeating the evaluation with epoch-correct geometry, that is, with the element set current on each night, raises the threshold-model gain under the quota from -0.2% to 9.2% and at a quarter budget from 55.0% to 75.9%, but lowers it at the half budget from 37.6% to 33.2%, the expected-value gain there being unchanged at 19.0%. Back-propagated geometry

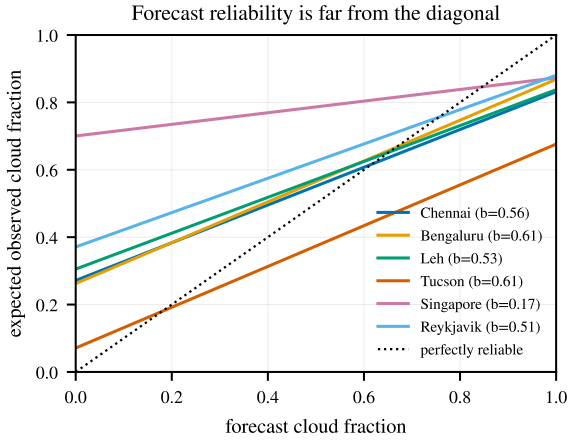


Fig. 4. Fitted forecast reliability. A perfectly reliable forecast would lie on the dotted diagonal; the fitted slopes are far below one, so the raw model field over-states its own discrimination.

therefore does not inflate the result systematically, although it is not uniformly conservative.

E. Conflict resolution in practice

On 2000 randomised instances small enough to enumerate exhaustively, the dynamic program matched the exhaustive optimum in 100.00% of cases. Greedy selection *by the score itself* attains 99.95% of the optimum on average, so the dynamic program offers little average-case improvement over it; its advantage is that it is exact, has negligible cost, and exhibits no adverse case. The rules an operator typically applies perform markedly worse (Fig. 5), since they optimise a proxy rather than the objective: selection by highest culmination attains 95.39% on average, is optimal in only 62.7% of instances, and falls to 9.4% in the worst case generated.

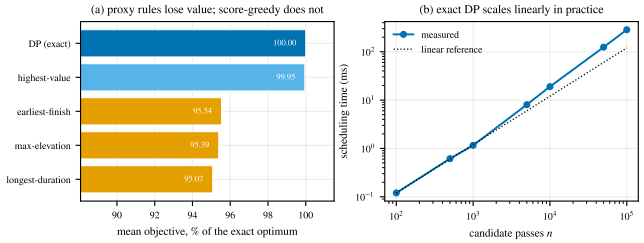


Fig. 5. (a) Greedy heuristics attain less than the exact dynamic program. (b) Scheduling time grows linearly in practice and remains negligible at catalogue scale.

On real candidate sets, an order-based genetic algorithm of the type common in the satellite-tasking literature (population 60, 120 generations) attains 99.67% of the optimum but requires approximately 38 922× more time, a ratio that depends on that search budget. Once the problem is recognised as interval scheduling, an exact polynomial solution is available and a metaheuristic is unnecessary. Scheduling is in any case not the bottleneck, the dynamic program handling 100 000 candidates in 285.3 ms against the seconds to minutes required for propagation.

F. End-to-end behaviour

The funnel is steep and physically interpretable (Fig. 6): at the demonstration site, 12 449 geometric passes reduce to 8093 sunlit, 1297 with the observer in darkness, 367 bright enough to detect, 87 above the score floor once forecast cloud is weighted in, and 26 scheduled. A weather-blind tool would present the operator with the earlier, substantially larger counts. Per-station runs also provide a physical consistency check: Svalbard (78.2° N) returns the largest number of geometric and sunlit passes of any station, yet *zero* optically observable passes, since in late August the Sun does not fall below the −6° dark-sky threshold there.

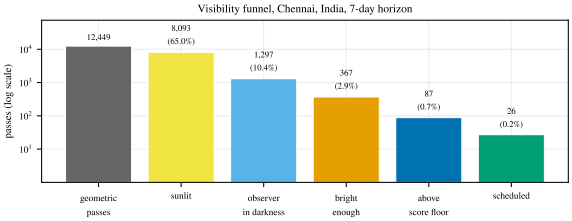


Fig. 6. Visibility funnel for the demonstration station: of 12 449 geometric passes over 7 days, 26 survive to the timetable. Cloud enters the score as a continuous weight rather than a hard cut, so the penultimate stage is the score floor.

Planning 635 objects over 7 days takes 109.4 s single-threaded, dominated by propagation, which scales linearly in objects and horizon. The timetable is also stable against the score weights: varying w_e from 0.3 to 0.7 leaves a Jaccard overlap of at least 0.89 with the default timetable, falling to 0.70 once elevation dominates ($w_e \geq 0.9$) and to 0.86 with duration alone. Within the moderate range the physical constraints, not the weighting, govern the outcome.

VI. Discussion and Limitations

Cloud cover is an imperfect proxy for observing conditions: total cloud fraction on a reanalysis grid cell of order 25 km describes a line of sight only coarsely, and transparency, seeing, cloud height and aerosol are not modelled. A local all-sky camera could supply a better nowcast, to which the calibration of Section III-C would apply unchanged, and sub-hourly skill would let a quota-constrained planner discriminate within a night. Neither value model captures partial credit; a station with a specific instrument would be better served by its own value function, which the scheduler accepts unchanged.

The gains pool six stations and 60 nights in one season. Their 95% intervals come from a bootstrap that resamples whole stations (10 000 draws); with so few clusters they are approximate and probably too narrow, and the calibrated-minus-raw difference was not tested separately.

We assume one instrument observing one object at a time; multiple apertures or a station network break the interval-scheduling structure and would require a different algorithm. Photometry is approximate, since (2) models every object as a diffuse sphere, so the magnitude gate is an ordering rather than a prediction. The ageing measurement is single-season

and restricted to low Earth orbit, where drag-driven decay dominates.

VII. Conclusion

SkyPass integrates pass prediction, optical visibility analysis, cloud-forecast reasoning and exact conflict resolution into a single planner suitable for deployment at a small ground station, producing a timetable that exports directly to a calendar or a rotator controller. Pass extraction agrees with an independent implementation to 0.055 s while requiring $23.7\times$ fewer propagator calls than dense sampling, and conflict resolution is exact at negligible computational cost.

Two weather-related findings are of particular relevance to the design of such systems. First, a forecast does not by itself improve performance: it is informative about which *nights* merit observation and much less so for ranking passes within a night, where cloud is autocorrelated and pass quality varies substantially more than sky conditions. A planner under a fixed nightly quota cannot exploit that information and gains nothing, whereas given a budget it may allocate selectively the same forecast yields 19.0% (expected value) and 37.6% (clear-sky count). Second, calibration against reanalysis raises the gain in every regime tested, and under a quota it is the difference between a loss and break-even. Both findings should carry over to other small facilities, though only one setting is tested here: the value of a forecast is a property of the decision it informs. The implementation and experiment scripts are released under the MIT licence.

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